

Min-Cut

Definition: A *multi-graph* is a graph $G = (V, E)$ where $E \subseteq V \times V \times \mathbb{N}$

Definition: A *cut* is a set of edges $C \subseteq E$ such that G without C is disconnected.

A *minimum cut* in G is a cut in G that has the minimum number of edges.

Problem Definition:

Given a multi-graph G find a minimum-cut in G

Min-Cut

Karger's Algorithm:

Input: multi-graph G

Output: min-cut in G

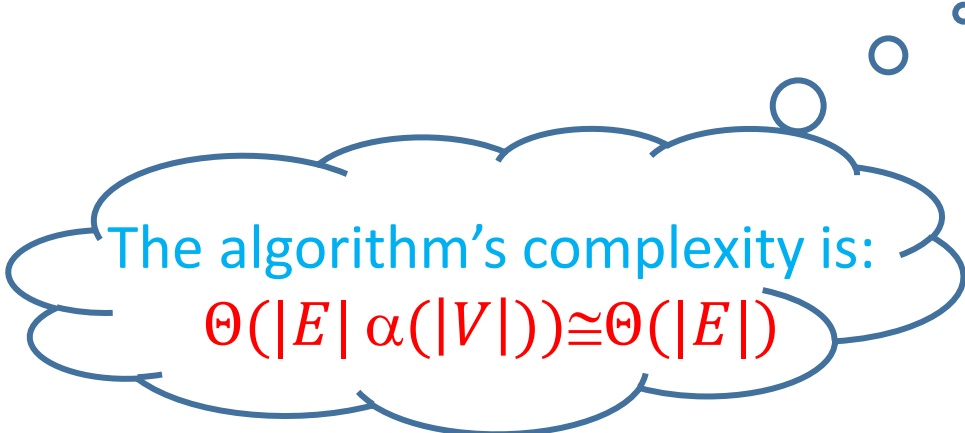
1. **While** the number of vertices in the graph > 2 , **do**:
 - a. Choose an edge (u,v) of the graph uniformly at random
 - b. Unite the vertices u,v
2. **Return** the remaining edges as the min-cut.

Min-Cut

Karger's Algorithm:

Input: multi-graph G

Output: min-cut in G

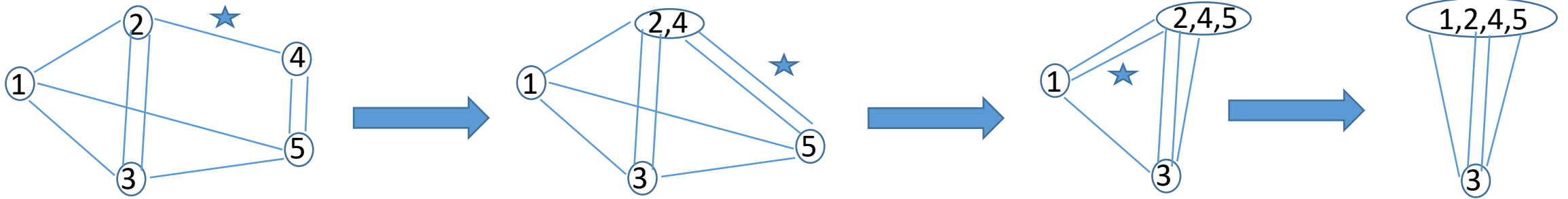


The algorithm's complexity is:

$$\Theta(|E| \alpha(|V|)) \cong \Theta(|E|)$$

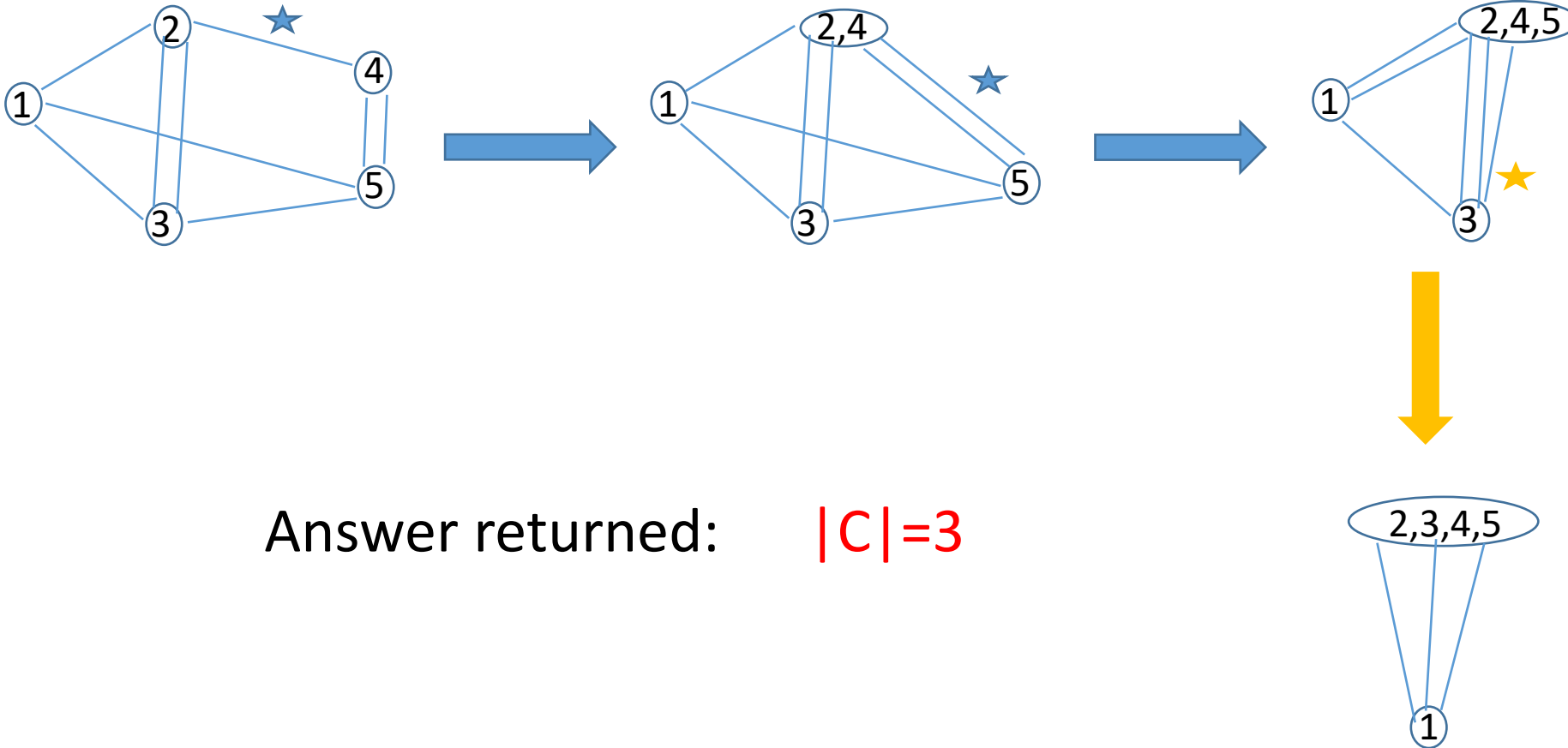
1. **While** the number of vertices in the graph > 2 , **do**:
 - a. Choose an edge (u,v) of the graph uniformly at random
 - b. Unite the vertices u,v
2. **Return** the remaining edges as the min-cut.

Karger's Algorithm for Min-Cut



Answer returned: $|C|=4$

Karger's Algorithm for Min-Cut



Karger's Algorithm for Min-Cut: Analysis

Denote by $G^{(i)}$ the graph obtained by the algorithm after iteration i , where $G = G^{(0)}$.

Claim 1: The algorithm returns a cut in the input graph G .

Proof: Let x and y be the last vertices remained in $G^{(n-2)}$.

Denote by:

$A_y = \{\text{The set of vertices in } G \text{ united to } y\}$

$A_x = \{\text{The set of vertices in } G \text{ united to } x\}$

Note that there are no edges in G between A_y and A_x that are not returned by the algorithm. Therefore, the set of edges returned by the algorithm is indeed a cut.

Karger's Algorithm for Min-Cut: Analysis

Claim 2: For all $0 \leq i \leq n - 2$ the size of the minimum cut in $G^{(i)}$ $\leq$ the size of min-cut in $G^{(i+1)}$.

Proof:

Let C be a minimum cut in $G^{(i+1)}$.

Note that C is also a cut in $G^{(i)}$ since cuts may be deleted in an iteration but not added.

Therefore, the minimum cut in $G^{(i)}$ also takes C into account.

The claim then follows.

Karger's Algorithm for Min-Cut: Analysis

Let K be the size of a min-cut in G , and let C be a cut of size K .

Claim 3: For all $0 \leq i \leq n - 2$ the number of edges in $G^{(i)} \geq \frac{K(n-i)}{2}$.

Proof:

Since the min-cut size is K , by claim 2 we have that the size of the min-cut in $G^{(i)} \geq K$, for all $0 \leq i \leq n - 2$.

Therefore, the degree of each vertex in $G^{(i)}$ is at least K . (Why?)

Also, the number of vertices in $G^{(i)}$ is $n - i$.

The claim then follows. (Why?)

Karger's Algorithm for Min-Cut: Analysis

We now want to compute the probability that the algorithm returns a correct answer (i.e. the algorithm does not return an error).

Observation:

The algorithm returns a correct answer if it returns C

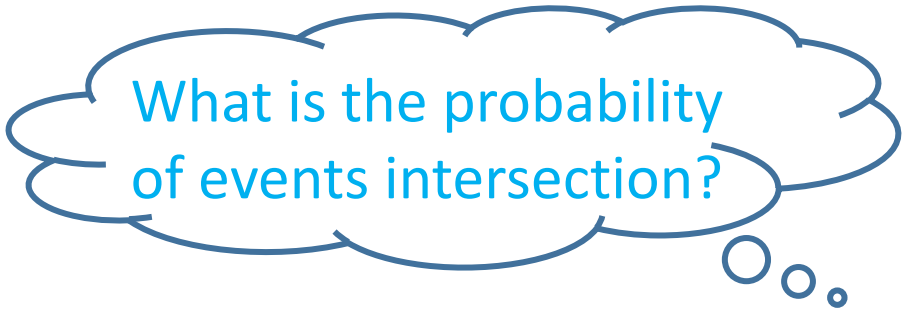
(Note that if C is not the only min-cut, the algorithm may return a correct answer even if it does not return C).

Therefore, the probability that the algorithm returns a **correct** answer is **at least** the probability that it never chooses an edge from C .

Karger's Algorithm for Min-Cut: Analysis

Denote by E_i the event that in the i -th iteration the algorithm did not choose an edge from C , for $0 \leq i \leq n - 3$.

$$P[\text{algorithm is correct}] \geq P \left[\bigcap_{i=0}^{n-3} E_i \right]$$



What is the probability of events intersection?

Basic Probability - Continued

Definition: Let E_1, E_2 be events. We say that E_1 and E_2 are *independent* if:

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$$

Note: For general events E_1 and E_2 it holds that:

$$P(E_1 \cap E_2) = P(E_1) \cdot P(E_2/E_1)$$

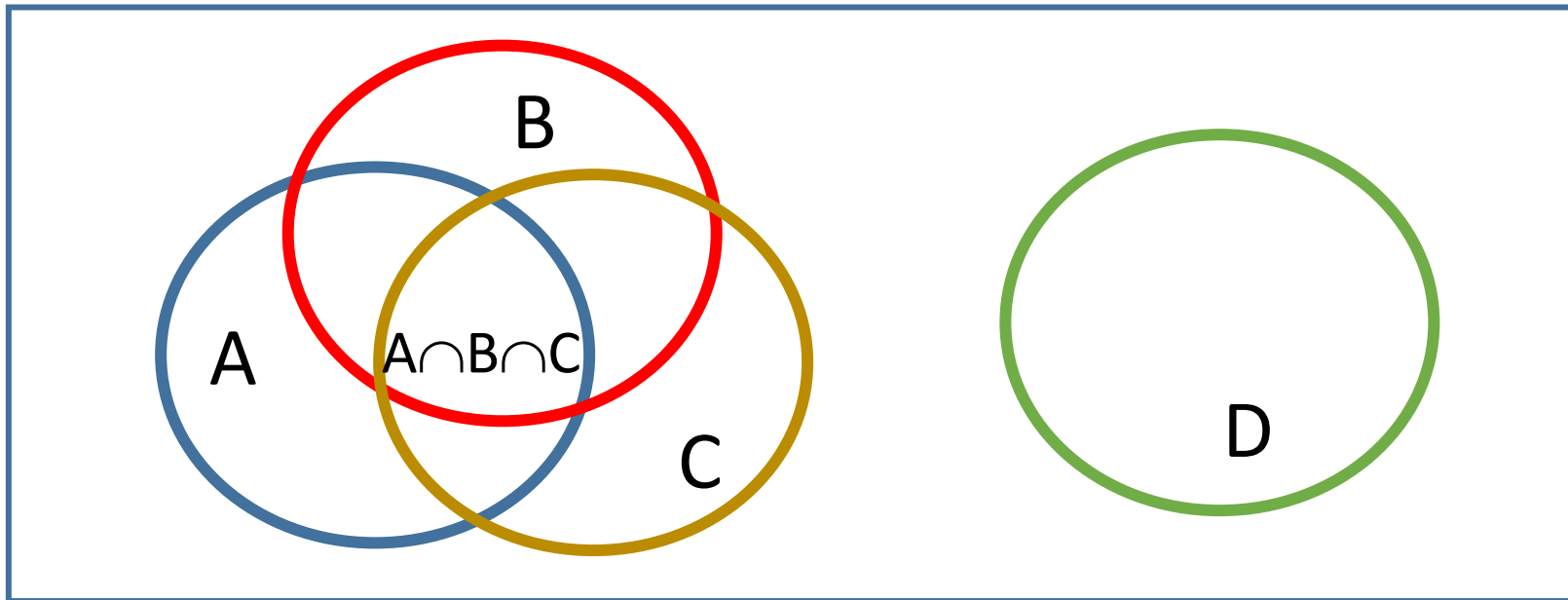
where:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Basic Probability - Continued

For a sequence of events $E_1, E_2, \dots, E_k$ it holds that:

$$P\left(\bigcap_{i=1}^k E_i\right) = P(E_1) \cdot P(E_2/E_1) \cdot P(E_3/E_1 \cap E_2) \cdots P(E_k/\bigcap_{i=1}^{k-1} E_i)$$



Karger's Algorithm for Min-Cut: Analysis

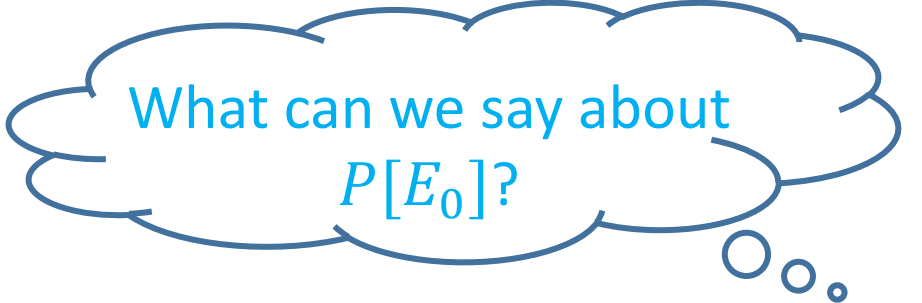
Denote by E_i the event that in the i -th iteration the algorithm did not choose an edge from C , for $0 \leq i \leq n - 3$.

$$\begin{aligned} P[\text{algorithm is correct}] &\geq P\left[\bigcap_{i=0}^{n-3} E_i\right] \\ &= P(E_0) \cdot P(E_1/E_0) \cdot P(E_2/E_0 \cap E_1) \cdots P(E_{n-3} / \bigcap_{i=0}^{n-4} E_i) \end{aligned}$$



Karger's Algorithm for Min-Cut: Analysis

$$P(E_0) \geq ?$$

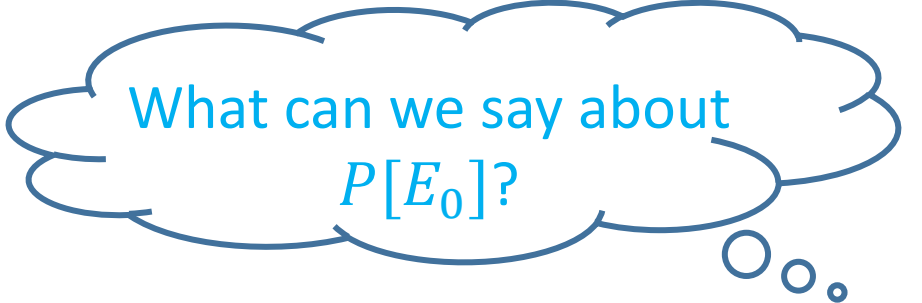


What can we say about
 $P[E_0]$?

Karger's Algorithm for Min-Cut: Analysis

$$P(E_0) \geq ?$$

The number of edges in $G = G^{(0)} \geq \frac{n \cdot K}{2}$ (Claim 3)



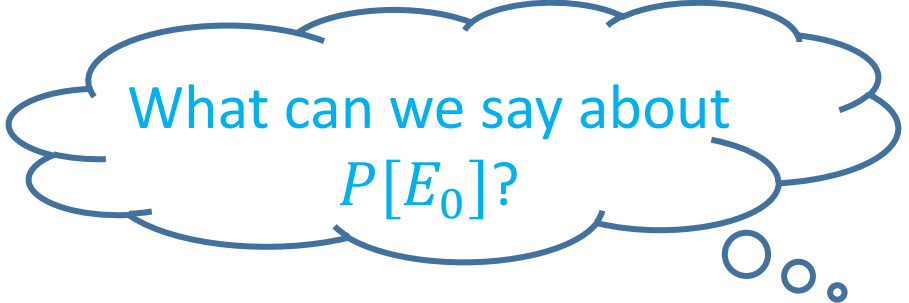
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Karger's Algorithm for Min-Cut: Analysis

$$P(E_0) \geq ?$$

The number of edges in $G = G^{(0)} \geq \frac{n \cdot K}{2}$ (Claim 3)

The probability of a random choice of an edge in $G^{(0)} \leq \frac{1}{\frac{n \cdot K}{2}} = \frac{2}{n \cdot K}$



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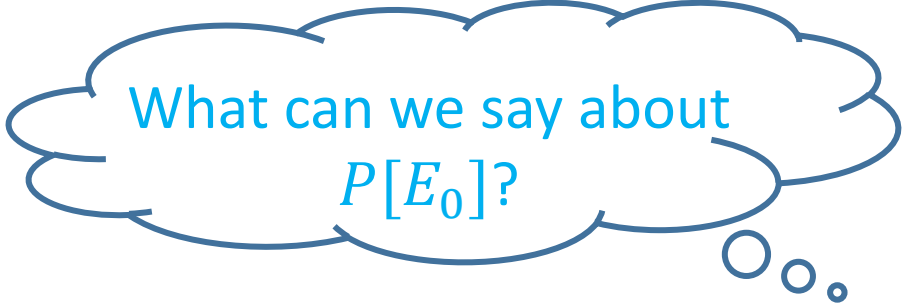
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The probability of a random choice of an edge **from C** in $G^{(0)} \leq \frac{K}{\frac{n \cdot K}{2}} = \frac{2 \cdot K}{n \cdot K} = \frac{2}{n}$



What can we say about
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Karger's Algorithm for Min-Cut: Analysis

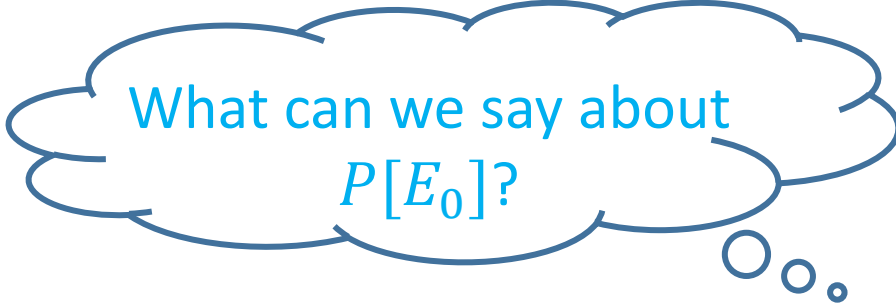
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The probability of a random choice of an edge **from C** in $G^{(0)} \leq \frac{K}{\frac{n \cdot K}{2}} = \frac{2 \cdot K}{n \cdot K} = \frac{2}{n}$

Conclusion: $P(E_0) \geq 1 - \frac{2}{n}$



What can we say about $P[E_0]$?

Karger's Algorithm for Min-Cut: Analysis

Claim 4: $P(E_i / \bigcap_{j=0}^{i-1} E_j) \geq 1 - \frac{2}{n-i}$

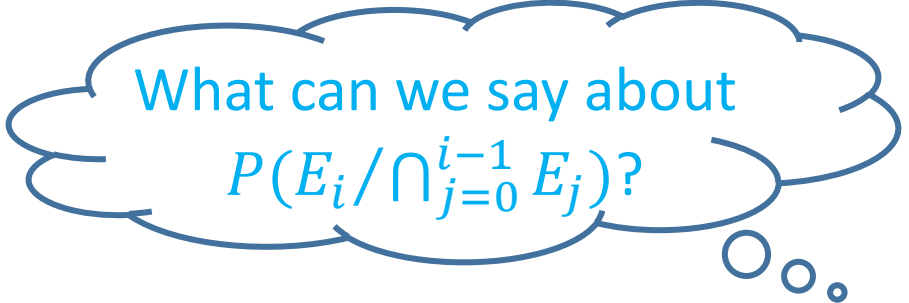
Proof:

The number of edges in $G^{(i)} \geq \frac{(n-i) \cdot K}{2}$ (Claim 3)

The probability of a random choice of an edge in $G^{(i)} \leq \frac{1}{\frac{(n-i) \cdot K}{2}} = \frac{2}{(n-i) \cdot K}$

The probability of a random choice of an edge **from C** in $G^{(i)} \leq \frac{K}{\frac{(n-i) \cdot K}{2}} = \frac{2 \cdot K}{(n-i) \cdot K} = \frac{2}{(n-i)}$

Therefore, $P(E_i / \bigcap_{j=0}^{i-1} E_j) \geq 1 - \frac{2}{n-i}$



What can we say about
 $P(E_i / \bigcap_{j=0}^{i-1} E_j)$?

Three small circles (ellipses) follow the bubble.

Karger's Algorithm for Min-Cut: Analysis

Conclusion: $P[\text{algorithm is correct}] \geq$

$$\begin{aligned} &\geq P\left[\bigcap_{i=0}^{n-3} E_i\right] = P(E_0) \cdot P(E_1/E_0) \cdot P(E_2/E_0 \cap E_1) \cdots P(E_{n-3}/\bigcap_{i=0}^{n-4} E_i) \\ &\geq \left(1 - \frac{2}{n}\right) \cdot \left(1 - \frac{2}{n-1}\right) \cdot \left(1 - \frac{2}{n-2}\right) \cdots \left(1 - \frac{2}{n-(n-3)}\right) \\ &= \prod_{i=0}^{n-3} \left(1 - \frac{2}{n-i}\right) = \prod_{i=0}^{n-3} \left(\frac{n-i-2}{n-i}\right) = \end{aligned}$$

Karger's Algorithm for Min-Cut: Analysis

Conclusion: $P[\text{algorithm is correct}] \geq$

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Karger's Algorithm for Min-Cut: Analysis

Conclusion: $P[\text{algorithm is correct}] \geq$

$$\begin{aligned}
 &\geq P\left[\bigcap_{i=0}^{n-3} E_i\right] = P(E_0) \cdot P(E_1/E_0) \cdot P(E_2/E_0 \cap E_1) \cdots P(E_{n-3}/\bigcap_{i=0}^{n-4} E_i) \\
 &\geq \left(1 - \frac{2}{n}\right) \cdot \left(1 - \frac{2}{n-1}\right) \cdot \left(1 - \frac{2}{n-2}\right) \cdots \left(1 - \frac{2}{n-(n-3)}\right) \\
 &= \prod_{i=0}^{n-3} \left(1 - \frac{2}{n-i}\right) = \prod_{i=0}^{n-3} \left(\frac{n-i-2}{n-i}\right) = \frac{n-2}{n} \cdot \frac{n-3}{n-1} \cdot \frac{n-4}{n-2} \cdot \frac{n-5}{n-3} \cdots \frac{4}{6} \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} = \frac{2}{n(n-1)}
 \end{aligned}$$

Karger's Algorithm for Min-Cut: Analysis

Conclusion: $P[\text{algorithm is correct}] \geq$

$$\begin{aligned} &\geq P\left[\bigcap_{i=0}^{n-3} E_i\right] = P(E_0) \cdot P(E_1/E_0) \cdot P(E_2/E_0 \cap E_1) \cdots P(E_{n-3}/\bigcap_{i=0}^{n-4} E_i) \\ &\geq \left(1 - \frac{2}{n}\right) \cdot \left(1 - \frac{2}{n-1}\right) \cdot \left(1 - \frac{2}{n-2}\right) \cdots \left(1 - \frac{2}{n-(n-3)}\right) \\ &= \prod_{i=0}^{n-3} \left(1 - \frac{2}{n-i}\right) = \prod_{i=0}^{n-3} \left(\frac{n-i-2}{n-i}\right) = \frac{n-2}{n} \cdot \frac{n-3}{n-1} \cdot \frac{n-4}{n-2} \cdot \frac{n-5}{n-3} \cdots \frac{4}{6} \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} = \frac{2}{n(n-1)} \end{aligned}$$

Is the probability good enough?

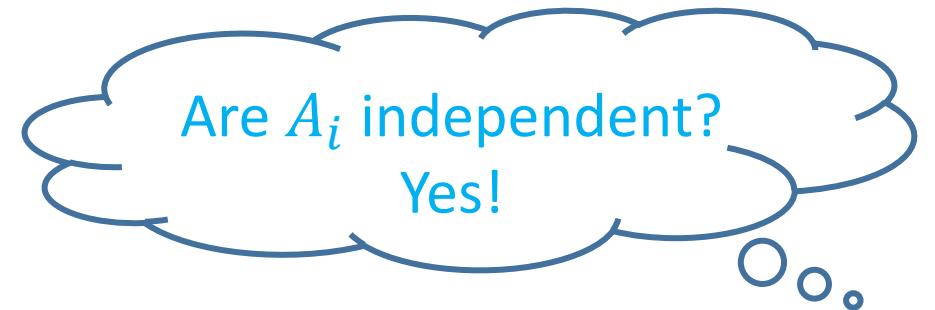
Probability Amplification

We will amplify the probability to get a correct answer:

*Repeat the basic algorithm X times and return the **minimum** result of all runs.*

Denote by A_i the event that in the i -th run the algorithm is wrong, $1 \leq i \leq X$.

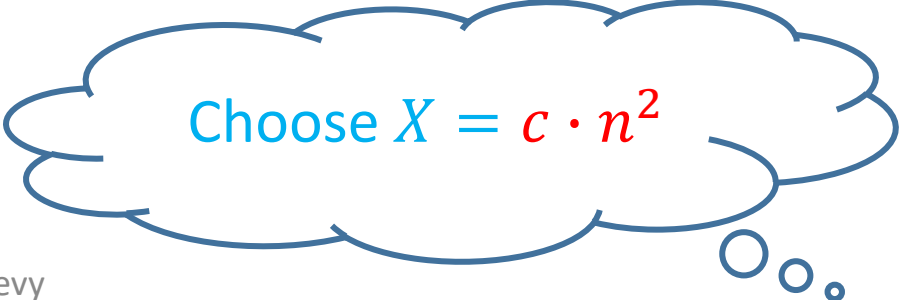
$$P[A_i] \leq 1 - \frac{2}{n(n-1)} \leq 1 - \frac{1}{n^2}$$



Probability Amplification

Repeat the basic algorithm X times and return the *minimum* result of all runs.

Denote by A_i the event that in the i -th run the algorithm is wrong, $1 \leq i \leq X$.

$$\begin{aligned} P[\text{amplified_algorithm is wrong}] &= P\left[\bigcap_{i=1}^X A_i\right] = \prod_{i=1}^X P[A_i] \\ &\leq \prod_{i=1}^X \left(1 - \frac{1}{n^2}\right) = \left(1 - \frac{1}{n^2}\right)^X = \left(1 - \frac{1}{n^2}\right)^{c \cdot n^2} = \left[\left(1 - \frac{1}{n^2}\right)^{n^2}\right]^c \leq [e^{-1}]^c \\ &= e^{-c} \end{aligned}$$


Choose $X = c \cdot n^2$

Amplified Algorithm's Complexity

Repeat the basic algorithm X times and return the *minimum* result of all runs.

The algorithm's complexity
is: $\Theta(|V|^2 |E| \alpha(|V|))$

Choose $X = c \cdot n^2$

Is it competitive?